Lab 7 — Capstone check-ins

Friday, December 4 launch · Monday, December 7 studio

Last updated: October 8, 2026.

Dates are on the schedule. The graded write-up and gallery live in the capstone packet. This lab page is the **in-class effort** track: short computational checkpoints and instructor initials.

Credit: Lab points are for effort in class, not for a Canvas file upload on this lab assignment. Work in Octave or Python. On check-in days you use a **self-contained one-pager** with space for answers and new syntax; get instructor initials after a short interaction. Preview the one-pager; write on the hard copy handed out in class. The **capstone PDF** is a separate Canvas upload (25 points in the Capstone group).

Resources: Lachniet Ch. 5 (SVD), Ch. 7 (image compression) · Syntax sheets · Resources · Capstone · Lecture 40.

What we are doing

Practice the SVD / storage / error moves you need for Plan A vs Plan B (independent frames vs F_1, D_2, D_3), then continue the full comparison in studio.

Learning objectives

1. Compute a small SVD and a rank-k reconstruction.
2. Count full vs rank-k storage.
3. Measure a relative reconstruction error.
4. State a capstone question in one sentence (accuracy / storage / runtime).
5. Make progress on Plan A vs Plan B with recorded numbers.

Math connections

SVD and rank-k ideas from Lecture 40; cost awareness from the micro-lab and Lecture 29.

Key syntax (this lab)

Task	Octave	Python
SVD	`[U,S,V] = svd(A)`	`U, S, Vt = np.linalg.svd(A, full_matrices=False)`
Rank-k rebuild	truncate U, S, V	`U[:,:k] @ np.diag(S[:k]) @ Vt[:k,:]`
Frobenius norm	`norm(A,'fro')`	`np.linalg.norm(A,'fro')`
Timing	`tic` / `toc`	`time.perf_counter()`

Exercises (launch / studio checkpoints)

Pick Octave or Python. You may use a tiny synthetic "frame" if you do not yet have photos.

E1. Let

$$A = \begin{bmatrix} 4 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

(or a small image patch). Compute the SVD. Form the rank-1 approximation A_1. Report σ_1 and $\|A - A_1\|_F$.

Expected: Depends on A; for the matrix above, $\sigma_1 = 4$ and the rank-1 error involves the second singular direction (hand-check or print norms).

E2. Storage. For $m = 100$, $n = 80$, $k = 5$: compute mn and $k(m + n + 1)$. Which is smaller?

Expected: $mn = 8000$; $k(m + n + 1) = 5 \cdot 181 = 905$.

E3. Differences. From three small matrices F_1, F_2, F_3, form $D_2 = F_2 - F_1$ and $D_3 = F_3 - F_2$. Verify $F_2 = F_1 + D_2$ and $F_3 = F_2 + D_3$ in code.

Expected: residuals numerically zero.

E4. Capstone sentence. Write one question you will answer with numbers (e.g. "At what k does Plan B match Plan A's error with less storage on my frames?").

E5 (studio). Run Plan A and Plan B at two values of k. Record two error numbers and two storage totals. (Full plots go in the capstone PDF.)

One-pager

Preview the Lab 7 ticket; write on the hard copy in class.